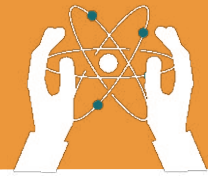


UNIT 7

ENERGY AND ITS FORMS

Learning objectives:

- Understand the forms of energy, including kinetic, potential, and heat energy.
- Explain the law of conservation of energy and its applications in energy transformations.
- Identify renewable and non-renewable energy sources and their uses.



Have you ever wondered why you feel energized after breakfast?

Food provides energy, allowing you to study, play, and do chores throughout the day. By evening, feeling tired means you've used up much of your energy and need rest.

What is Energy?

Energy is the ability to do work. It is measured in joules (J). Energy exists all around us in various forms and is essential for performing tasks.

Examples of Energy Sources

- The heat and light from the sun
- Wind currents
- Fuel used in vehicles
- Electricity that powers our homes

Potential Energy (P.E.)

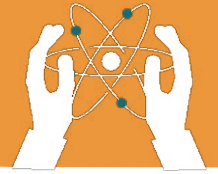
Stored form of energy is considered as potential energy. There are two types of potential energy.

1. Gravitational Potential Energy:

An object at rest on the ground has no potential energy. However, when lifted to a certain height, work is done against gravity, and energy is stored as **gravitational potential energy**.

Example: A brick lifted off the ground possesses gravitational potential energy

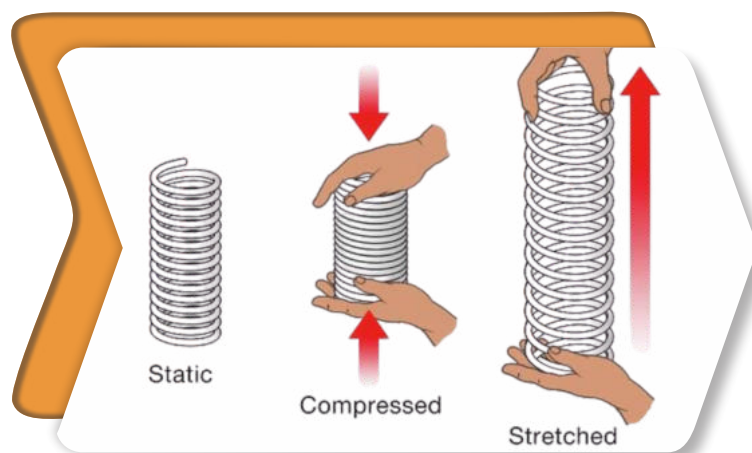




Elastic Potential Energy

Certain materials can be stretched or compressed and will return to their original shape. The energy stored in such materials is called **elastic potential energy**.

Example: A stretched rubber band or a wound spring



Key Concepts

Energy: The ability to do work.

Kinetic Energy: Energy due to the motion of an object.

Potential Energy: Energy due to the position of an object.

Energy Converter:

A device that converts one form of energy into another.

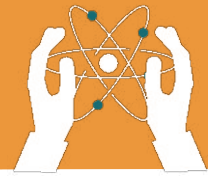
Solar Panel: A device that converts solar energy into electricity.

Activity: Elastic Potential Energy Demonstration

Objective: Show elastic potential energy.

Materials: Rubber bands.

Instructions: Stretch and release rubber bands to observe stored energy release.

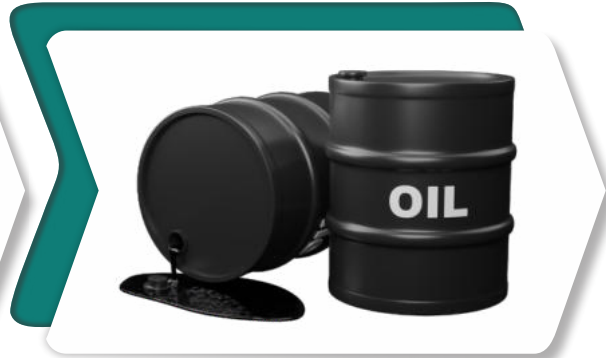
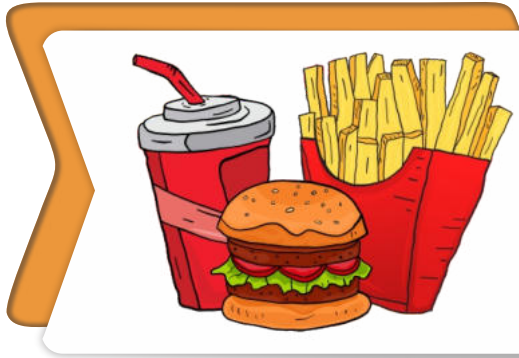


ENERGY AND ITS FORMS

Chemical Energy

This is the energy stored in substances due to chemical processes, such as in food, batteries, and fuels.

Example: Energy stored in wood, oil, and batteries



Kinetic Energy (K.E.)

Kinetic energy is the type of energy an object has due to its motion. Every moving object has kinetic energy.

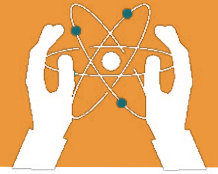
Examples:

- A flying airplane
- A boy cycling
- A moving car



Heat Energy

The energy of a substance due to the movement of its particles is termed as heat energy. As a substance heats up, its particles move faster, increasing its temperature.



Energy as a Physical Quantity

Energy is expressed as work done and can be measured. The standard unit for energy is the Joule (J).

Law of Conservation of Energy

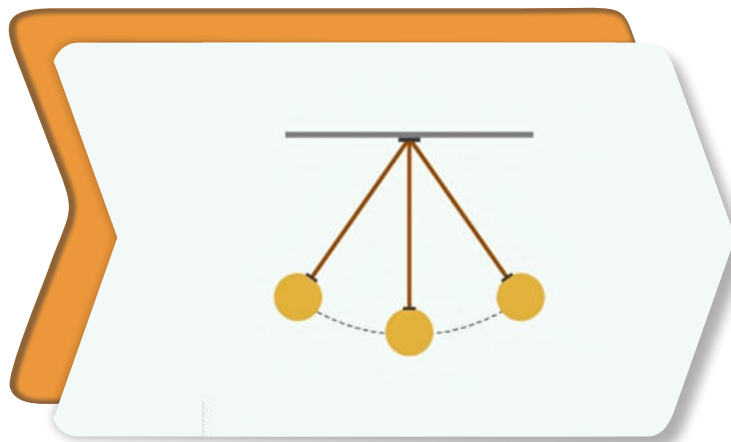
The **Law of Conservation of Energy** states that energy cannot be created or destroyed; it can only be transformed from one form to another.

Example: Simple Pendulum

As a pendulum swing:

At its highest points, it has maximum potential energy and no kinetic energy.

At its lowest point, it has maximum kinetic energy and no potential energy.





ENERGY AND ITS FORMS

Energy Converters

Devices that convert energy from one form to another are known as energy converters.

For example:

A washing machine converts electrical energy into kinetic energy.

A solar panel converts solar energy into electrical energy.



washing machine



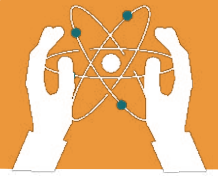
solar panel

Energy Conversion Experiment

Objective: Observe energy conversion.

Materials: Battery, light bulb, wires.

Instructions: Connect battery to bulb to demonstrate chemical energy converting to light and heat.



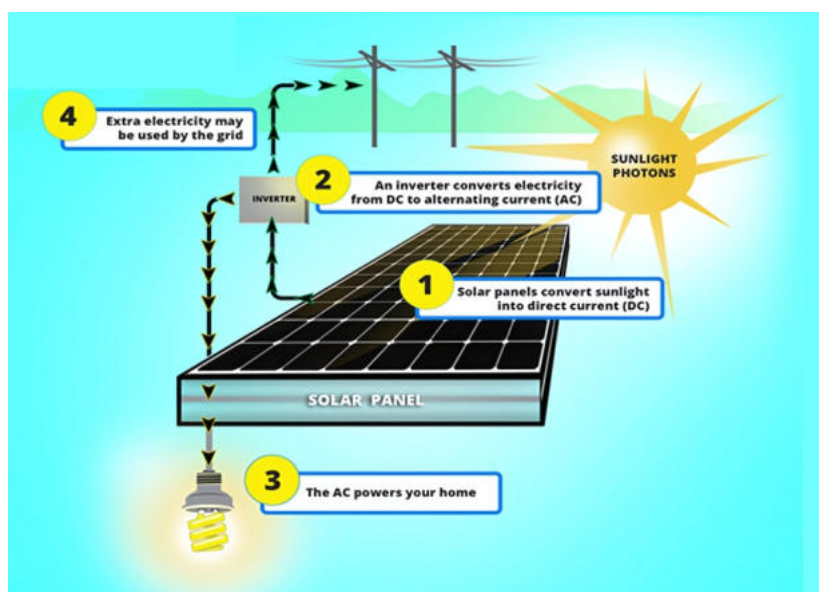
Sources of Energy

• Renewable Energy Sources

These are energy sources that can be replenished:

Solar Energy: Solar panels convert sunlight into electricity.

Wind Energy: Wind turbines convert wind energy into electrical energy.



• Non-Renewable Energy Sources

These sources are finite and cannot be replenished quickly:

- Coal
- Natural Gas
- Crude Oil



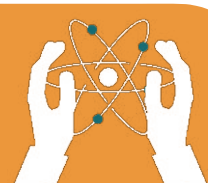
Coal



Crude Oil



Natural Gas



ENERGY AND ITS FORMS

Fill the blanks with the help of given words in the box.

joules (J)

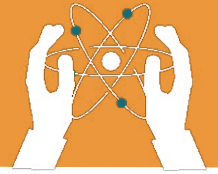
renewable

elastic

Conservation
of Energy

chemical

1. Energy is measured in _____.
2. A stretched rubber band has _____ potential energy.
3. Wind and solar power are examples of _____ energy sources.
4. The energy stored in food is known as _____ energy.
5. According to the Law of _____, energy cannot be created or destroyed, only transformed.



Choose the correct answer.

1. Which of the following is a source of renewable energy?

- Coal
 - Wind
 - Natural Gas
 - Petroleum
-

2. Which of the following devices is an example of an energy converter that changes solar energy into electrical energy?

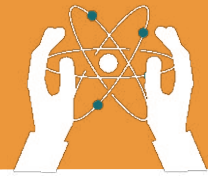
- Washing machine
 - Wind turbine
 - Solar panel
 - Generator
-

3. Which of the following is a non-renewable source of energy?

- Wind
 - Solar
 - Natural Gas
 - Hydropower
-

4. Which of the following is an example of chemical energy?

- Wind
 - Food
 - Solar power
 - X-ray
-



5. What kind of energy is primarily associated with moving objects?

- Potential Energy
 - Nuclear Energy
 - Kinetic Energy
 - Elastic Energy
-

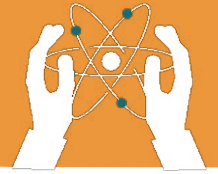


Critical Thinking

Energy Transformation:

When you ride a bicycle, multiple energy transformations take place. Explain these transformations and suggest ways to make cycling more energy efficient.

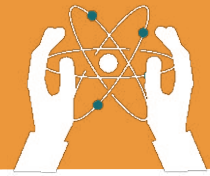
ENERGY AND ITS FORMS



The Right Energy

What kind of energy is produced with the following items? Write your answers in the space provided.

	Hammering a nail		Machine
	Petroleum		Coal
	Natural Gas		X-Ray
	Girl who is jogging		Nuclear Plant
	Space Exploration		Turning the radio on.
	A cup of hot chocolate		Typing in the keyboard
	Cooking fries		A boy running because he is late



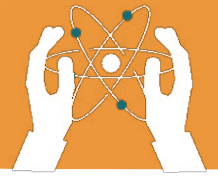
Answer the Following Questions:

Short-Answer Questions

- Q1. What is energy?
- Q2. Name four different types of energy.
- Q3. Explain the difference between kinetic and potential energy.
- Q4. What are energy converters? Also give an example.

Long-Answer Questions

- Q1. Explain the difference between kinetic energy and potential energy, providing examples of each.
- Q2. Describe the Law of Conservation of Energy and provide an example to illustrate this concept.
- Q3. Differentiate between renewable and non-renewable energy sources with examples.



Perform the experiments and give reasons:

1. A metal spoon and a wooden spoon are kept in hot water for five minutes. The metal spoon feels much hotter than the wooden spoon. Why does this happen?

2. When a ball is dropped from a height, it bounces a few times but eventually stops. If energy is neither created nor destroyed, where does the energy go?

3. A person is holding a heavy bag without moving. Is work being done? Why or why not?

Project ideas

Energy Transfer Diagram: Create a diagram showing how energy changes from one form to another using everyday examples.

Solar Panel Project: Assemble a small solar panel to power a fan.

Comparison Chart: Compare renewable and non-renewable energy sources and their advantages.